

Yes, I am. I'll already go for it. Well, thank you for allowing me to be in your space. I'm really excited, and I'm sorry I can be there today, a little under the weather, so it's safe for me to stay home. But I'm excited to work with you this semester as we think about a prototype. We are very interested in an electronic medical record. Um, I would hope they're vaguely somewhat familiar with what that might look like, either through your experience, going to a doctor's office or within the hospital. But what I envision, it's a military discipline type of system. So it would encompass both capabilities for inpatient, or what we call acute care, and outpatient, which I also call ambulatory care. So while you're in the hospital and while you're not, that's kind of how I see those 2 things. I also mention it being multi disciplined, so for example, those who can access would be like family medicine, psychiatry, physical therapy, so being multiple disciplines would be using the medical record in both the settings. And what I've seen done is that in the medical record, when you log in, there'll be a town for, like, the acute care side, which is a hospital, and then there'll be a top tap for the ambulatory care site. Or which would be the outpatient side. So as I mentioned, there would be some roles, um, even admin, position nurses, front desk, uh, the builder, and a patient portal that's optional, not sure if we can get to that point. There's also. Permission levels. So, in a medical record, usually, like, psychiatry has more sensitive acts notes, so they have stricter access, as to who can see those notes. Um, there'd be a capability capability of audit money, like who acts as who changed what and when. Um, for the patient themselves, uh, as you're seeing someone who's using the medical record, uh, we would get registration information from them, or demographics, insurance information, emergency contact information. We should be able to look up, search and look up a patient, or duplicate a merge, different profiles. There would be a patient status, outpatient versus currently admitted, but if you have just the login for inpatient login for outpatient that kind of takes care of that, So forth, let me see, there should be a provider directory, so if you pull down, it'll have a list of some of the providers. I can also list disciplines or credentials in their schedules. as well as there's some discipline-specific note templates. So it's what we call a soap, S O A P. It's subjective objective, assessment, and plan, for general med, and then different structure for the different type of disciplines. There'd be an ability to send out referrals, uh, in both places, that way that you'll refer to the provider they need to get provided to. For clinical documentation, um, that we would have the encounter or the visit note, and it'd be one per visit, uh, and then they could, for inpatient, there's ongoing notes, right? it's a different setting. They have like progress notes or nursing shift notes or multiple entries per day. There also be a medical history, allergies, a problem list, immunization, history. Then there's vitals, that would be a single snapshot for the outpatient side versus frequent time series tracking with inpatient. For the inpatient only, you would have an admission work full, workflow, a bed unit room assignment and transfers, a discharge, summary generation, a care team tracking, multiple active providers and nurses per patient. And then for both, there be the ability to order labs, get results by uploading imaging orders and results by uploading referral tracking. And for medications, which is it's dear to my heart. Is outpatient having a current medication list, a prescription record, being able to prescribe as having

prescription authority. Inpatient, there'd be a medication administration record or what we call them R. It tracks each actual dose given by whom when any allergy or interaction checking. A nice feature would also be to have pharmacogenomics. And also to check for drug interactions. And see or duplications where it gives you an alert. That there there's a duplication of the drug or it gives you an alert that it's maybe overdosed too much. From ecogenomics is when we test their blood genetically, they may have a particular gene type that tells us more information about how the drug's going to be processed, so it works really fast, which doesn't work at all. So that capability would be great. Scheduling, outpatient, appointment based calendar for provider, and patient bed unit occupant CPU, not appointment based. Room resource scheduling participant specific needs. For example, like PT equipment, then there'd be a feature for billing an insurance. That's optional, or it could be very simplified, because we don't work too much on that, but our students, all of us should be able to know how to build for that service. So we should at least know what type of insurance they have a basic charge generation. What's important to include in the billing and insurance is the CPT or ICD code. I believe we're on ICD 10 and that's a manual that can be completely uploaded for us to select the ICD code based on what they have or what the problem is. For reporting an analytics, the volume per provider or discipline, an inpatient census, an occupancy reports. And then basic export like a CSV or PDF profile. Um, Let me see, it should be for security, we should have it, uh, HIPAA compliant. Of course, it's just an educational thing. So it doesn't need to be certified. For data integrity, we should have self delays or versioning on clinical records. never hard to lead. Some kind of encryption by trails. So we can identify, basically, the students are doing performance and visibility for physical, clinical staff, and then, um, I think that's probably like all we need for like safety. Also, it'd be great that students get assigned passwords. So I guess there's a manager that's kind of giving who's Evan, who gives access to the record. So the accounts are created based on their role because then that gives them kind of access they they need per discipline. Um, we could do a feature down in the bulk account creation from a class roster. We can add it already. People classes start to get an access. Each student's account is natural role for simulation purposes. Act D as a physician, nurse, or whatever the discipline is. Um, the system will generate a temporary password and then require the password to be changed on the 1st login. There's a password policy just to teach them these complexity with collecting a password. I know to use on this that is super important in our space. because we're always having to change almost every 7 days. Um, a self-service password. um Session timeout after inactivity, that's a real deal. And then um... Maybe some test patients so we could teach them how to use the system. Yeah, I think that's a lot. I try to get some of my feedback from some of my colleagues in the entition side, but I'm fully on the outpush inside, and I can definitely answer my questions on that case. Okay, perfect. So, does anyone have any questions? Yes. Can you briefly repeat the part you mentioned where patients will have like a qualifier that tells how the drug works on them? Oh, yeah. So, it's a medication component. So in most records, there'll be a tab for prescriptions, right? And you can see what they're actively on. And it has the feature of also being able to add, delete, or refill what

we call those drugs. In those, while we're prescribing to give them a drug, so let's say I'm going to give you moxycillin, it should have a quick drug interaction tracker to make sure that the amoxicillin isn't interacting with other drugs and make it ineffective. For example, if you have birth control, then it makes the antibiotic is not as effective, as effective, and that's something for detail patients. It should also include, so it's a drug interaction, if the dose is too high, if it's a duplicate, that means, oh, they already got another prescription from axis when it was given 2 days ago. It should have the ability, um, to, uh, oh, check for uh, pharmacogenomics type of data. So if I'm a formator, which I am, so I can use myself as an example, things like hydrocodone don't work on me because it never activates in my system. So the only drug that works on me is smoking, so that kind of inflammation prevents them from prescribing hydrocodone for me for pain management, for example. Um, and then it then, electronically, we can send out those prescriptions to a pharmacy. So there should be a list of all the different policies. In the area, you just kind of pull a listing of what's available, or it has the capability to print out a prescription for us design. Um, Yes, I don't I'm trying to think what else. I don't know if that answers the question. Yes. Okay, anyone else? Have any questions? I have a question. Do you know, like, new resources where we can find, you know, your, you say that this, like, um, platform should be able to, like, handle, um, you know, the, like, this drug that I'm taking right now is, um, like not working because of this other drug I'm taking. Yeah. Like, how do we know like where to find that? Yeah, it's back what the medical records use. If I heard you read, it's the resources. They'll use companies like Lexicon. Um, It's, of course, a subscription they have to put into the system. But I believe there's, I want to say, free resources out there. I can't think of what they are right now, but if you look at, you know, free resources like Lexicom or, um, facts and comparison, those are things that give you all of that and it gets integrated into the system so you don't have to manually add all that weird detail. But it'll that the manual is uploaded for functionality to be able to track all that. Very good. Any other questions? If something comes up, how would we reach out to? Okay, that's for me. Let me talk to you. No, I'll tell you how we're gonna, how we go about that. So for your password policy, you said that you guys have to have changed your password every 7 days? Yeah, but for the purpose of this, we don't have to do it. Okay. I was thinking more about that. As long as just once a temporary is given a reset is done. There's something to be said about the following instructions. Okay. We can even simulate that. Are there any features that require, um, additional, like, authentication? Like after login? Yes, we do have that feature. Um, It's kind of like duo. Actually, kind of is by duo. Yes, so there is a second verification step. It's awesome. Go ahead, work, work. Sorry, there's also a feature where people can dictate, but I'm not sure that we need that. Why not what? Sorry. Why not? We'll add it. I'm sorry, I didn't understand that. I said why not? We will have it. Well, you know, um... Oh, I'm saying that I'm not sure that we need it, because which part of me is like, well, we want to teach them how to write their notes. But you know what? If it's easy to add a dictation feature where there's a little mic and it dictates for you, then that's... Maybe then, for AI. And we'll set it up to disable it, if... Yes. It will be nice. I guess this is the, you know, I can pick whatever, like... Exactly. Do

you have any, like, documentation of, like, all this, like, all these things that you're speaking of? you just spoke up right now? Do I have documents? Like, did you like, like, like, have it in a dog for, um, like your requirements? Yeah, your requirements? No, this was her records. I don't know. She shared it with me, but I'm a female. All right, any other questions? Any questions about the system that you're developing? Going once. Yes. Is there like a, kind of like a, some example that might be out there that you could give us to kind of give ourselves a better idea of what exactly we're waiting for? You can probably Google to see the different types of EMR systems and ask for a demo. Because there's a lot out there. There's like centricity, epic. I don't think those are the best examples. Um, I forgot the one they texted, I just switched to, or switching to in 2027. So there's different formats. The educational platforms that are out there that we're currently paying for is EHR Go, and it's, we don't like it. And actually, you know, it's really using it, but as an instructor. So we're getting ready to remove that this year. Would you mind elaborating why you don't like that solution? Um, it's not very user friendly. It's multiplication. uh, there's multiple, uh, let's just say, I want to do one item and there's multiple versions of it and it's just really confusing. It doesn't feel like a real medical record. And then you have to create patient cases, and as soon as you use the patient case, and things are uploaded in there, but it doesn't, um, it does it's not as, it doesn't allow for flexibility. Yeah, you can save the cases, but if you have a truly a good, good running EMR system, you can create a test patient the students use. Would you be able to repeat the roles that you want for the system? I'm sorry? Would you be able to repeat the roles that you had in mind for this year? Sure. Well, I'm thinking of our UTEP health. So I'm thinking nursing, I'm thinking physical therapy, occupational therapy, social work, pharmacy. I'm thinking medical, right? That could be anything from family medicine, very different from internal medicine. And in the inpatient, there's a lot of disciplines, like infectious disease, physicians, oncology, physicians, surgeons, that has a whole different role set. But for the outside outpatient care, it's more than people that are seeing you for follow up, right? So physical therapy, occupational therapy, what else does each have health have? A speech language pathology, social work. We rehab sciences. I think we have something else. So I want to at least start with the rules that are hearing to help plus medical because it's important to have. Psych, psychology. And Admonds, right? Because people, there is a degree here for how to be a builder and administrator. But that could be a second phase, unless that's a easy part. Yeah, it's pretty easy. A front desk, that's another person, 'cause you're the ones who, when you arrive there, the the first people you see, and they're logging in information about you. Usually the medical records that I've seen is when I open it up, I'll have a nice, like, summary page. If so I pull up a patient, it'll give me like this nice summary, like, one little block of, like, the most recent visits, a little block that has, like, their demographics that I can see, a little different, uh, will block their labs, laboratories, and laboratories, too. That's an easy manual as well to add, so we just have to find the menu for lab. That's what we need to work with, is UTAP laboratory, because they have all those different labs and coding for it. And then that's kind of the interface and then you search between tasks to do different things. Good. Do you have any other questions? Or solid? This is a lot to take in for the

moment. Okay. Well, thank you so much, Dr. Pedia. We really, really appreciate you, and we'll be in touch with you soon. I hope you feel better. Oh, me too. Thank you, everyone. Thank you. Thank you. Thank you. All right. So that's our project. For what? Um... A little aside, the idea behind this is that we would actually develop something that would actually be used for educational purposes here at Utah. And the idea is that if this is really, really, really good, then we'll write a grant and we'll get money to pay for it after the back. But for now, we get free labor. So... Or, for now? We got this. So, really, ultimately, that's why she was saying you took health and all that, because it really is for you, Tim. Um, students. So pharmacy students, PT students, all those students be able to do this. And one of the most important things, as you guys will know, about anything having to do with medicine and all that, is documentation. So this is where we really have it. Okay, does this make sense? All right, ready to know your teams? All right, so we'll start with team number... Should I go like a backwards, right?? Tim number 4 will be Jasmine Huerta, Francisco Vasquez, Anaya Quinn, Derek Gamboya, Gamboya, excuse me, and Christian Rivilla. That will be team number four. Team, number three, will be Isaac Badia, Carina Rivera, Nestor, Escovedo, Guerdado, Orea, Fernando Munoz, that's team, number three. Team number two is Daniel Marin, Emmanuel Sainz, Jacob Zulva, Kindly, Wagmuel, and Zac Wimen, and team number one is Gladisa Dominguez, Jomota, Daniel Reyes, and Cerardosias. All right, so what we're gonna do, team number one, team number two, team number three, and team number four. So let's... How much? You know her? No. No. So you should, I should have taken your A or...